

StratosCore Rev A – Claude one-shot candidate (NOT FOR MANUFACTURE)

01 USB–C input

Arquivo: 01_usb_input.kicad_sch

02 2S battery, protection, charger

Arquivo: 02_battery_charger.kicad_sch

03 Rails

Arquivo: 03_rails.kicad_sch

04 Compute and control

Arquivo: 04_compute.kicad_sch

05 Display and touch

Arquivo: 05_display.kicad_sch

06 Sensors

Arquivo: 06_sensors.kicad_sch

07 GNSS

Arquivo: 07_gnss.kicad_sch

08 LoRa SX1262

Arquivo: 08_lora.kicad_sch

09 ADS–B receiver

Arquivo: 09_adsb.kicad_sch

10 Storage and audio

Arquivo: 10_storage_audio.kicad_sch

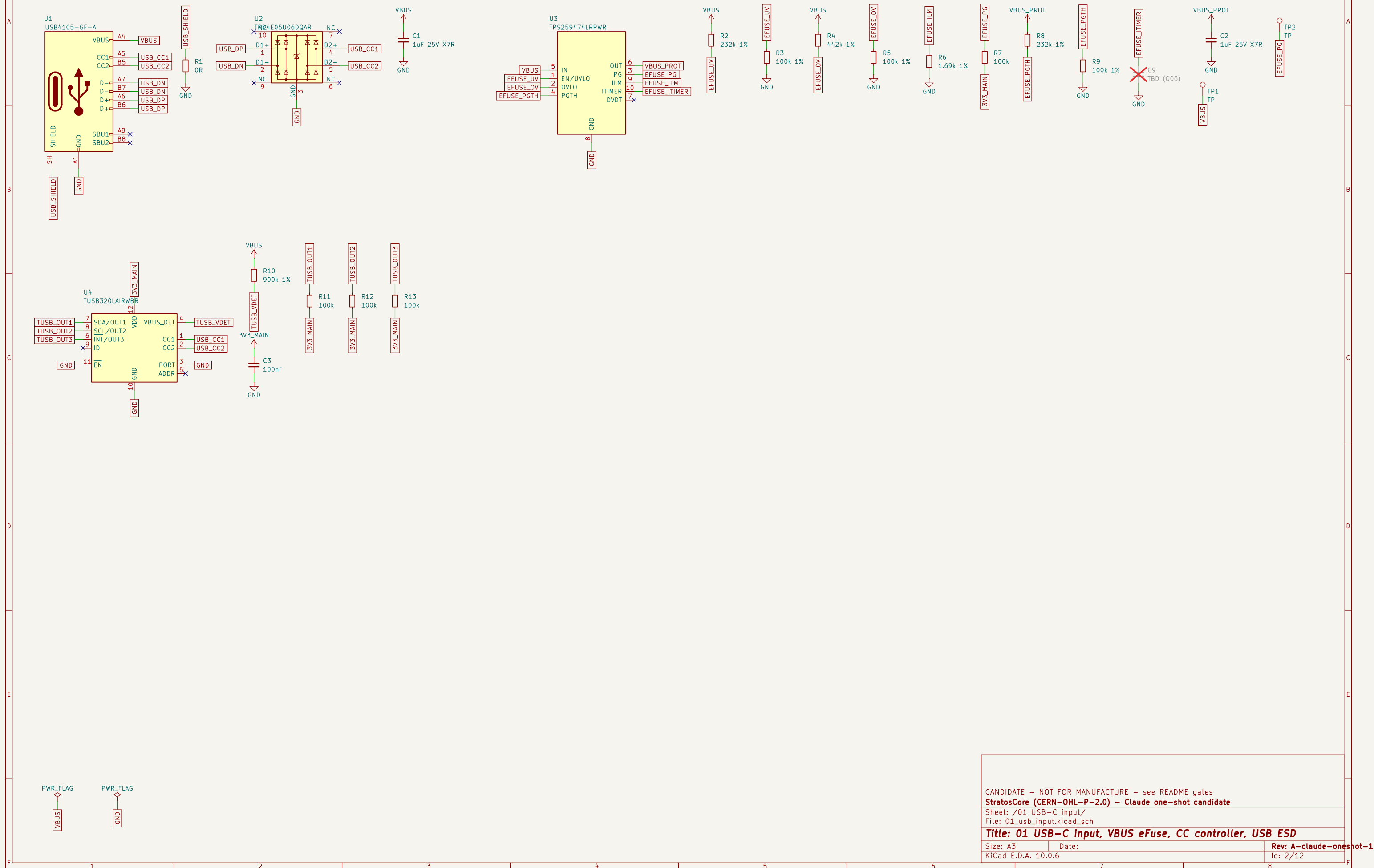
11 Mechanical

Arquivo: 11_mechanical.kicad_sch

Candidate produced for owner comparison with the Astra pass. docs/ASTRA_HANDOFF.md is NOT READY;
every open gate (005 battery safety, display controlled drawing/samples, RF values, assembler approvals) remains open.
Values marked TBD are deliberately not invented. See ../README.md and ../docs/ISSUES.md.

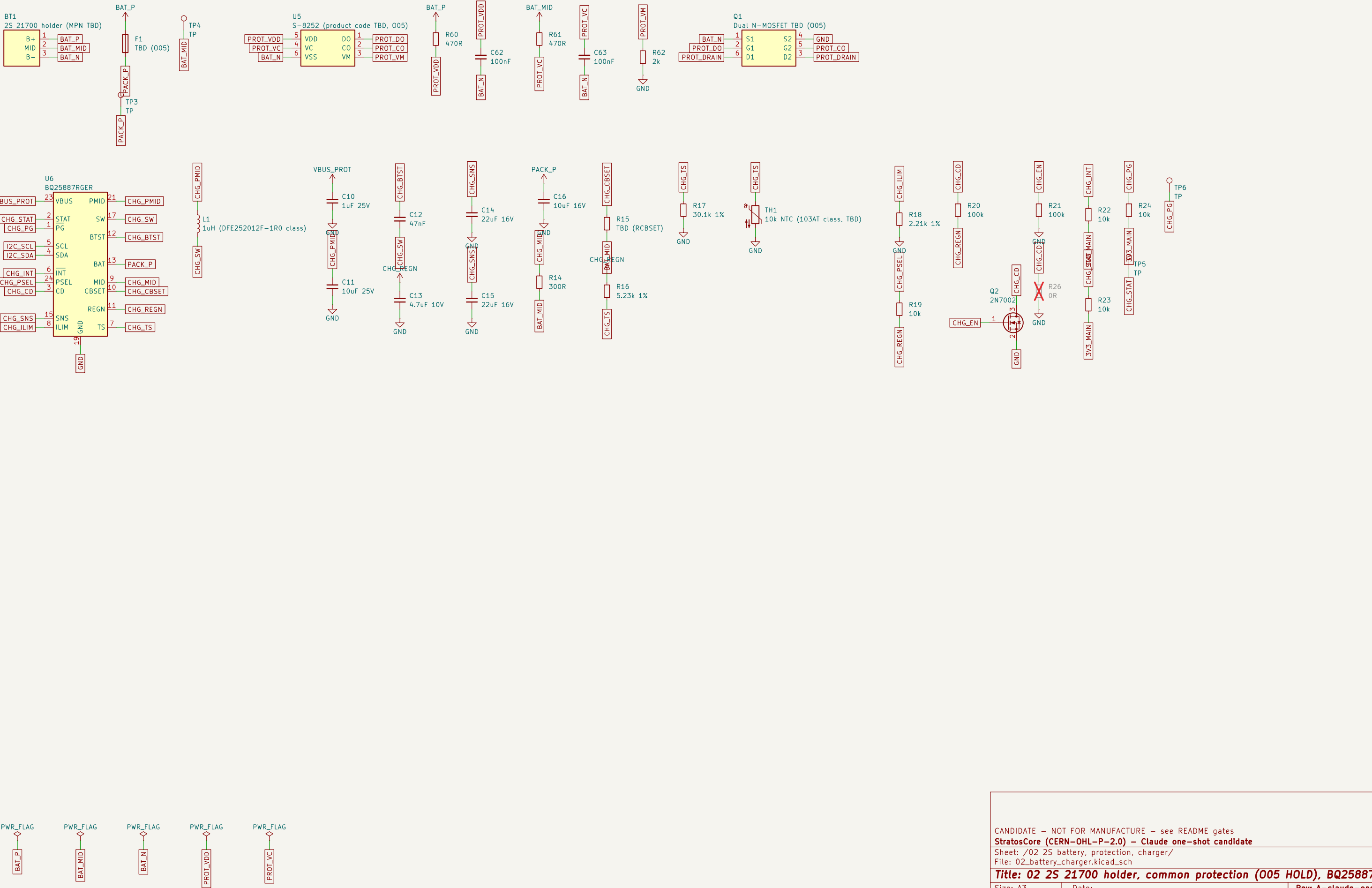
01 USB-C input, VBUS eFuse, CC controller, USB ESD

Fixed UFP sink, no USB PD, 5 V only (D2/P17). TUSB3201AI in GPIO mode (D22); PORT=GND, ADDR=NC, EN_N=GND, OUT1=3 read by TCA9535. TPS259474L values from docs/POWER_INPUT_EVIDENCE.md (PWR=02): RILM 1.69k, UVLO 232k/100k, UVLO 442k/100k, dVdI open, ITIMER TBD (006): C9 DNP. PGTH divider (R8/R9) is a candidate choice of this one-shot (same ratio as UVLO); review with O06. Shield treatment TBD: R1 OR placeholder.



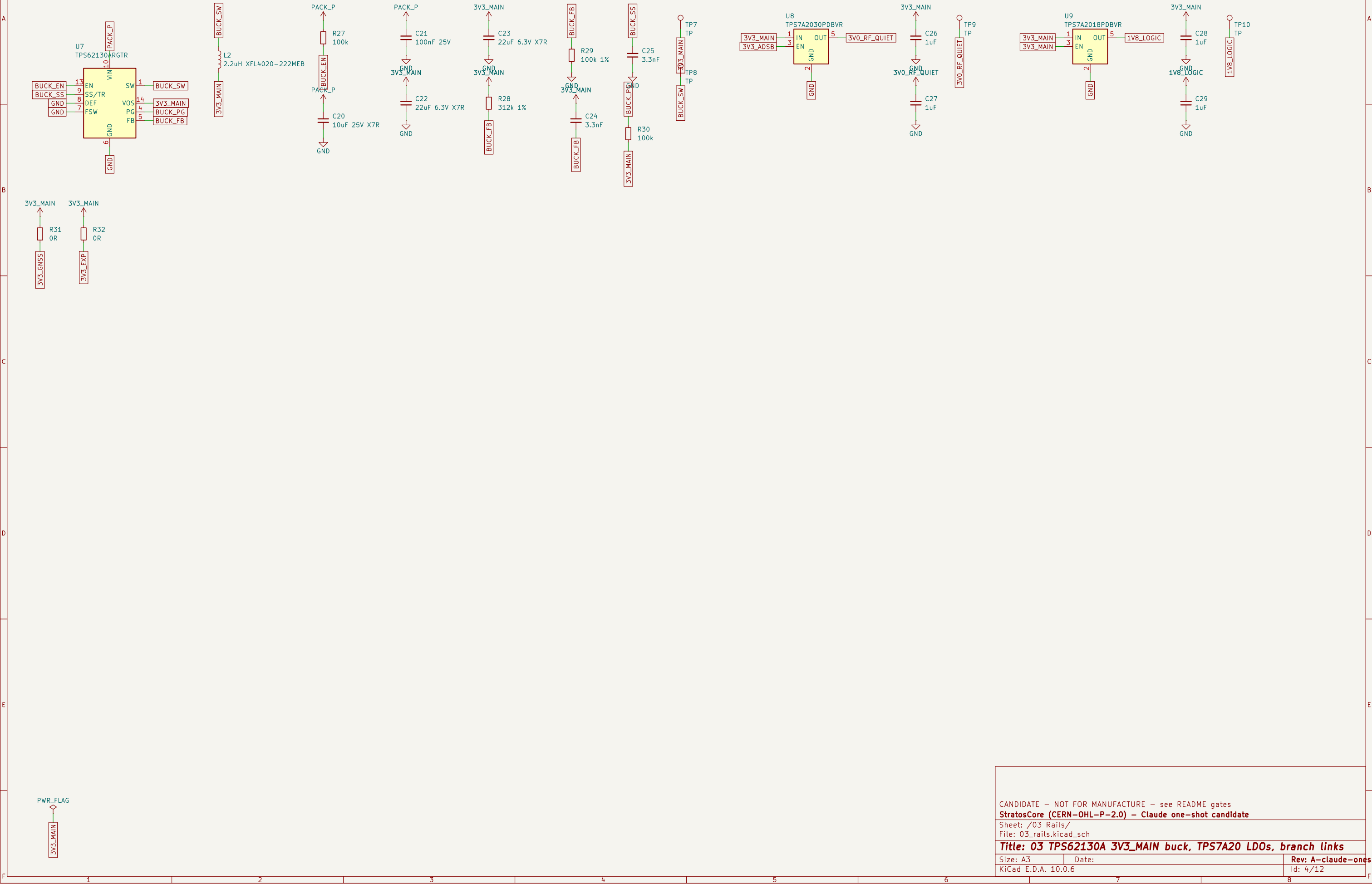
02 2S 21700 holder, common protection (005 HOLD), BQ25887 balanced charger

*** 005: battery protection/charger topology REQUIRES QUALIFIED HUMAN REVIEW. This sheet is a candidate only; do not energize. ***
Low-side common protection (S-8252 family + dual N-FET, product code/FETs/fuse TBD) is shown in the documented conflicting position: charger GND = pack negative after cutoff.
BQ25887 per SLUSD898 Fig.69: 1uH PMID-SW, 47nF BTST, 4.7uF REGN, 10uF PMID, 44uF SNS, 10uF BAT, TS 5.23k/30.1k + 10k NTC, 300R MID limiter.
Fail-safe CD: pulled to REGN (charging disabled whenever VBUS is present) until firmware drives CHG_EN (TCA9535 P17) -> Q2 pulls CD low. RILIM 2.21k -> 502 mA (KILIM 1110). PSEL high.
Dead-pack deadlock (MCU cannot enable charging if pack is empty) is an open 005 item; R26 (DNP) is a service override only.



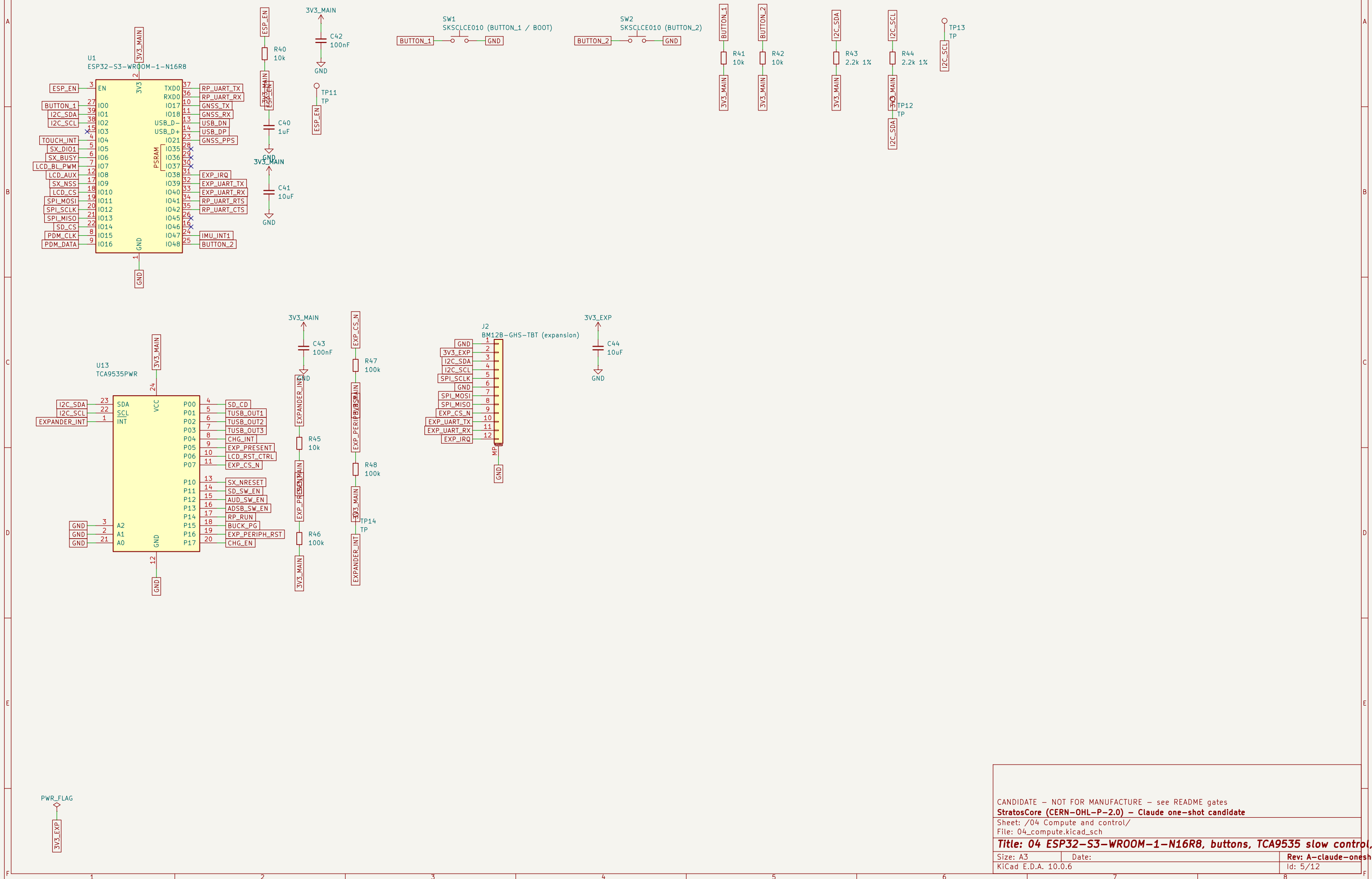
03 TPS62130A 3V3_MAIN buck, TPS7A20 LDOs, branch links

TPS62130A per SLVSAG7F 9.2: 2.2uH XFL4020–222MEB, FSW=GND (2.5 MHz), DEF=GND, 312k/100k -> 3.296 V (docs/POWER_RAIL_PLAN.md).
SS/TR: repo text lists 0.1uF (50 ms by Eq.10) – this candidate uses 3.3nF (about 1.65 ms, TI typical); feed-forward C24 3.3nF fitted per repo text. Reviewer to confirm.
No main power switch/button is specified in the product baseline: buck EN is pulled to PACK_P (always on while the protected pack is present). Product decision gap.
3V0_RF_QUIET LDO EN follows the switched 3V3_ADSB domain so the ADS–B analog chain powers down with RP2040 (one–shot proposal).



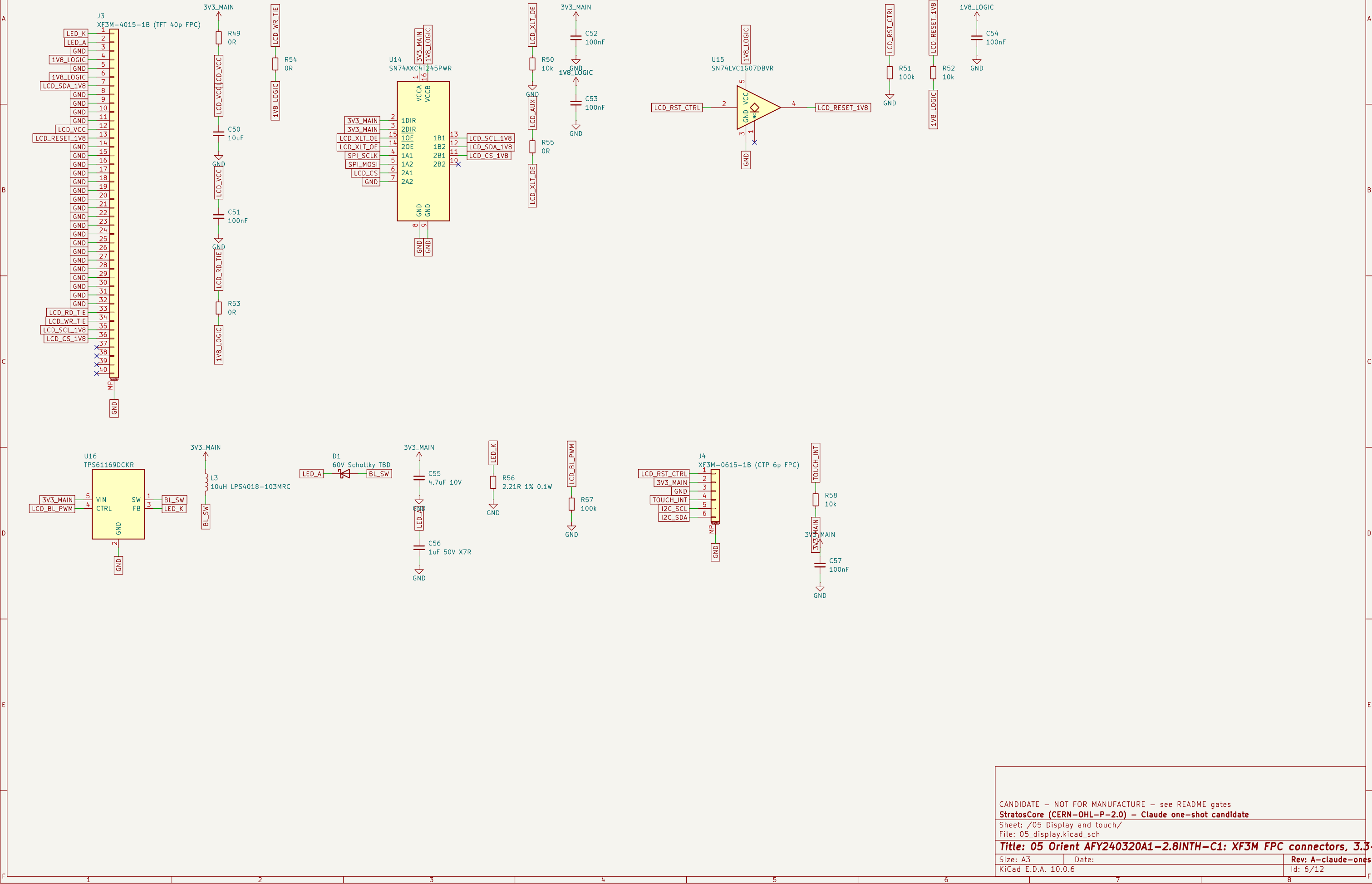
04 ESP32-S3-WROOM-1-N16R8, buttons, TCA9535 slow control, I2C pull-ups, expansion

GPIO allocation = docs/INTERFACE_GPIO_MAP.md. GPIO3/45/46 straps and GPIO35-37 (octal PSRAM) are no-connect.
 TCA9535 map = docs/CONTROL_IIO_REVIEW.md Dig-03 with ONE-SHOT PROPOSALS for the spare ports: P07 EXP_CS_N (expansion CS has no ESP32 GPIO in the map),
 P10_X_NRESET (was reserved SX aux), P14 RP_RUN (DIG-01 host reset), P15 BUCK_PG (input), P17 CHG_EN (charger enable, hardware default = disabled).
 I2C 100 kHz, 2.2k pull-ups on 3V3_MAIN (D22). JST GH BM12B-GHS-TBT is a VERTICAL (top-entry) header in JST/KiCad naming, not right-angle as the BOM text says.



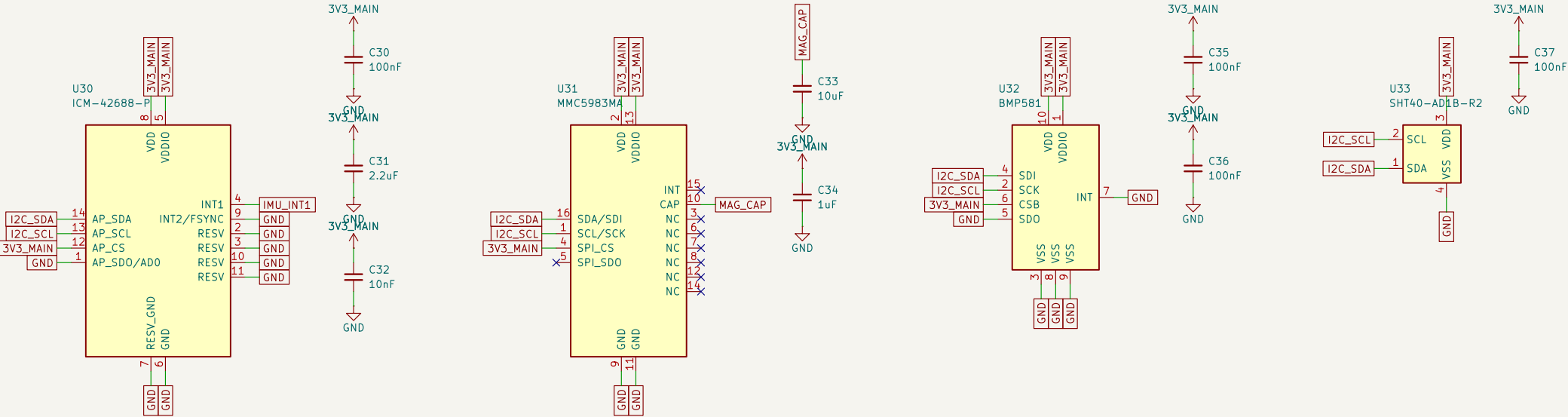
05 Orient AFY240320A1–2.8INTH–C1: XF3M FPC connectors, 3.3–>1.8 V SPI branch, reset, backlight

Write-only three-line 9-bit SPI, IM2/IM1/IM0 = 1/0/1 (rev J). SN74AXC4T245: DIR high (A->B), OE pulled low (enabled) and driven by GPIO8 LCD_AUX.
TFT reset: TCA9535 P06 LCD_RST_CTRL (100k pull-down) -> SN74LVC1G07 open drain -> 10k to 1V8_LOGIC (DSP-02). Touch RESET shares LCD_RST_CTRL (3.3 V, one-shot proposal).
UNUSED-PIN TIES ARE TBD (gate 3): DB17..DB0, DOTCLK/DE/VSYNC/HSYNC tied to GND; RD/WR tied high through R53/R54 OR (swap to GND if the controller spec requires).
TFT VCC fed from 3V3_MAIN (2.4–3.6 V range) – the VDD/VDDI contradiction in the Orient drawing is still open (001). Backlight TPS61169: RSET 2.21R -> -92 mA.



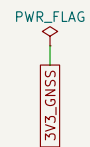
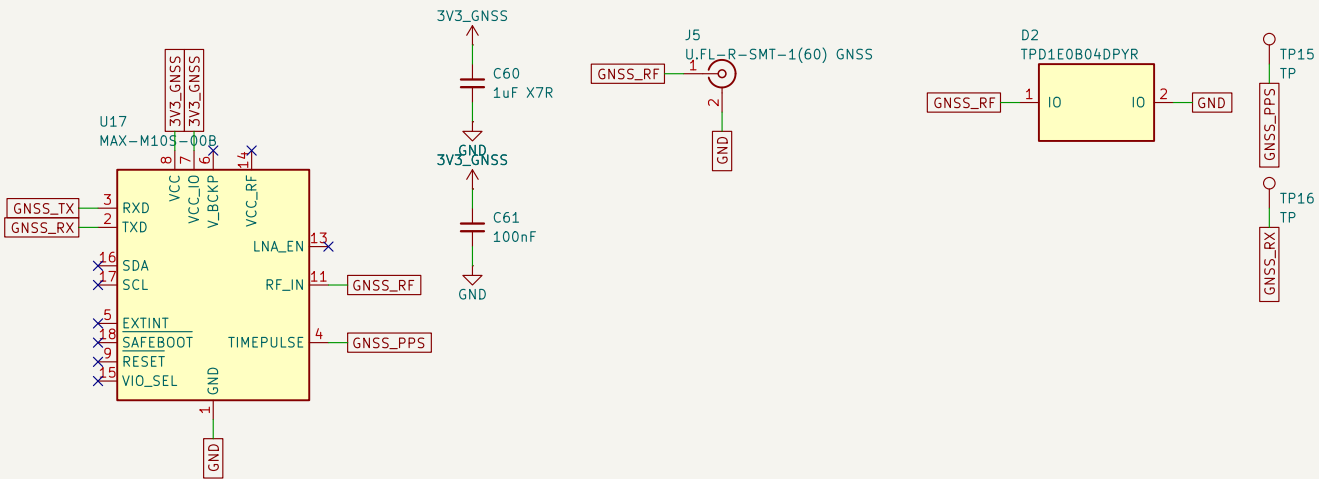
06 ICM-42688-P, MMC5983MA, BMP581, SHT40 on the shared 100 kHz I2C bus

Addresses: ICM-42688-P 0x68 (AD0=GND), MMC5983MA 0x30, BMP581 0x46 (SDO=GND, CSB=VDDIO), SHT40-AD1B 0x44, All on 3V3_MAIN.
Placement (SNS-02): IMU near centroid >=3 mm from anchors; MMC >=15 mm from cells/inductors; BMP581+SHT40 at the top vent edge, no copper/vias under BMP581, SHT40 on a slotted thermal tab.
BMP581 INT tied to GND with interrupts disabled (repo 04_sensors sheet); MMC INT unused (polled/NC); ICM pin 9 INT2/FSYNC/CLKIN tied to GND (TDK DS-000347 v1.9 Table 10: GND if FSYNC unused; PIN9_FUNCTION reset INT2 open-drain, keep disabled).



07 u-blox MAX-M10S-00B, passive antenna U.FL, TPD1E0B04 ESD

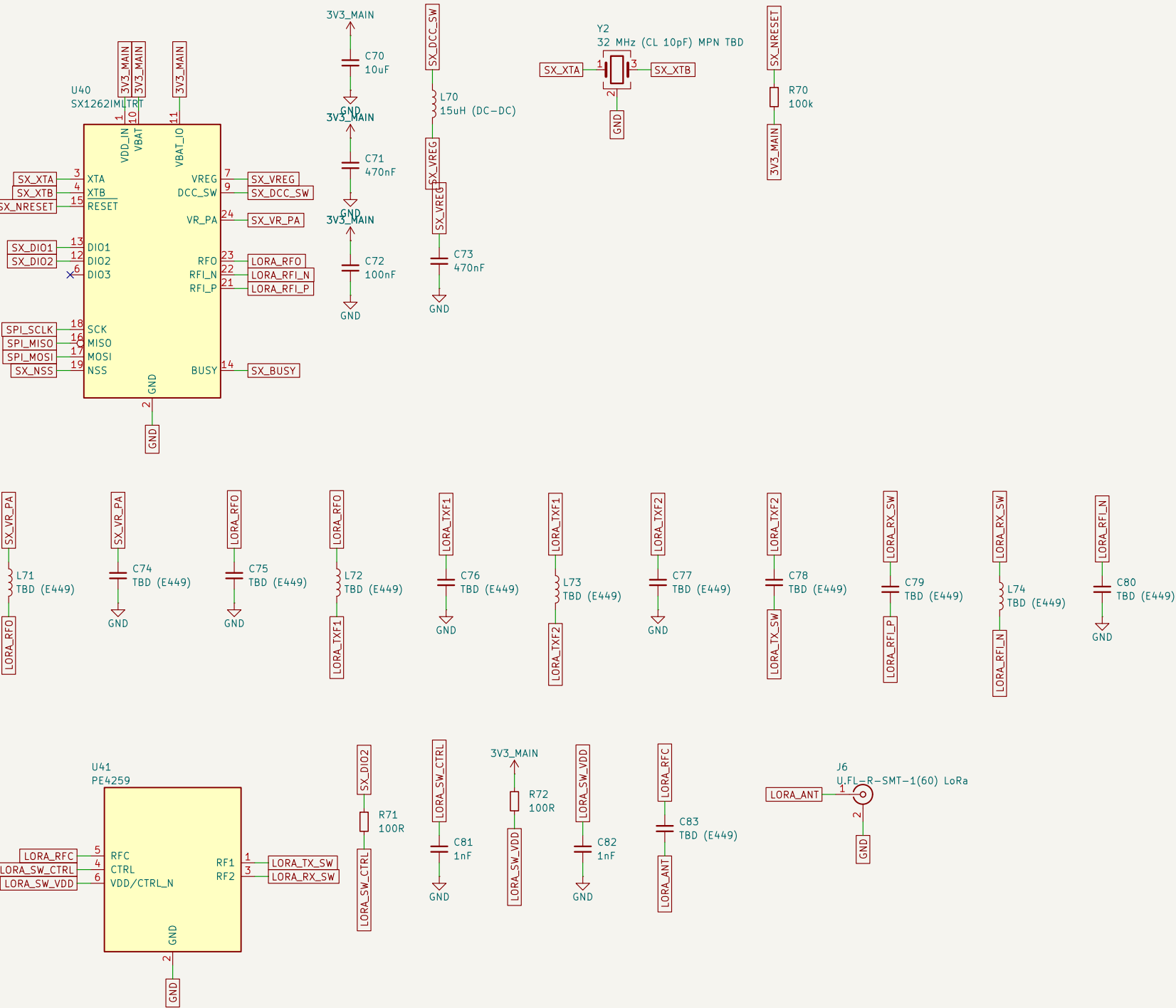
P20/GNSS-01: VCC and V_{IO} tied to 3V₃_GNSS, V_{IO_SEL} open, V_{BCKP}/RESET_N/EXTINT/LNA_EN/VCC_RF/SDA/SCL/SAFEBOOT_N open. RF_JN: short 50-ohm GCPW from U.FL (J5) with shunt TPD1E0B04 at the connector. No bias tee (passive FXP611 antenna). Firmware sets AIR4 dynamic model.



CANDIDATE – NOT FOR MANUFACTURE – see README gates		
StratosCore (CERN–OHL–P–2.0) – Claude one-shot candidate		
Sheet: /07 GNSS/		
File: 07_gnss.kicad_sch		
Title: 07 u-blox MAX–M10S–00B, passive antenna U.FL, TPD1E0B04 ESD		
Size: A3	Date:	Rev: A-claude-oneshot-1
KiCad E.D.A. 10.0.6		Id: 8/12

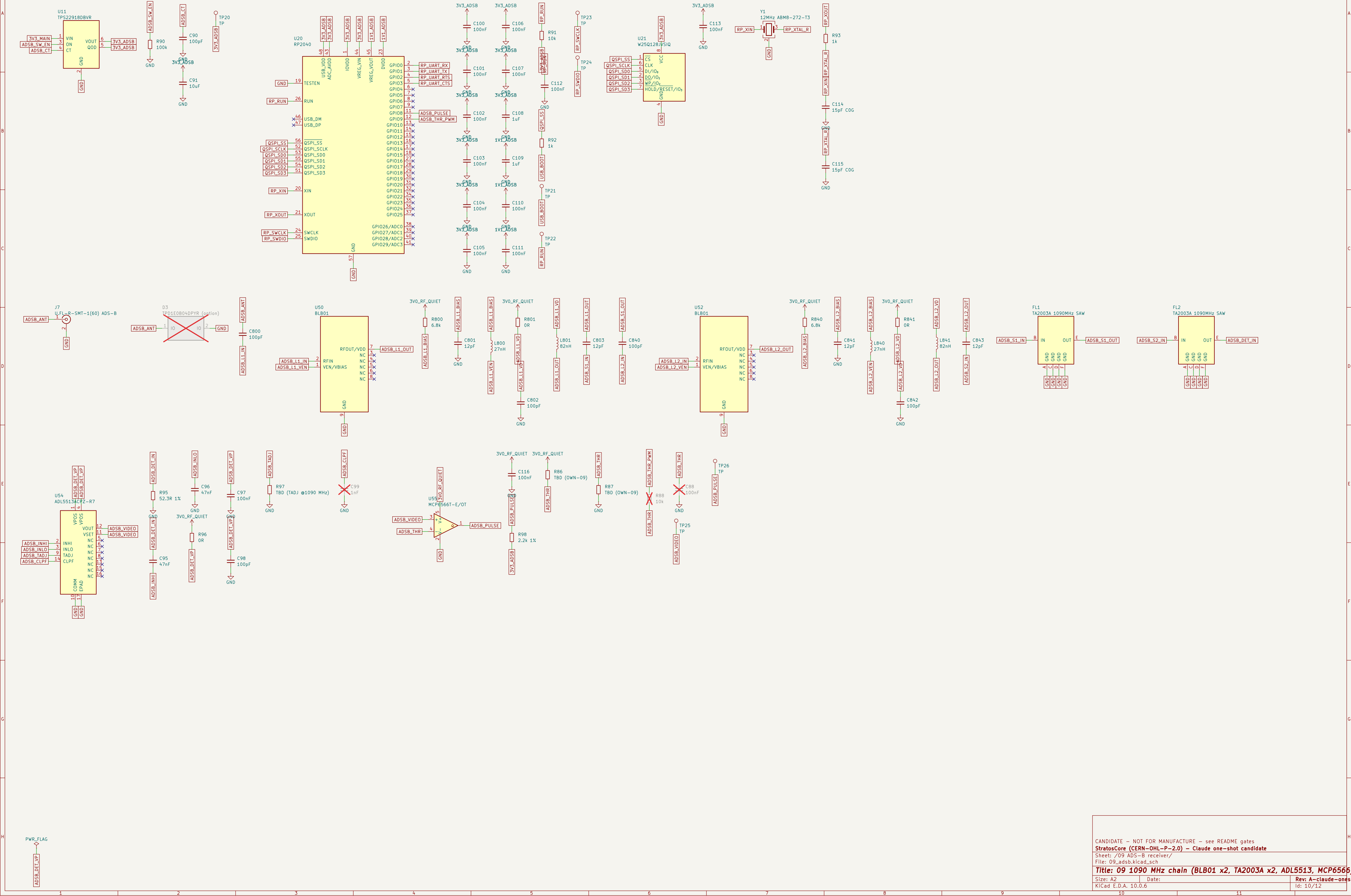
08 Semtech SX1262 (DC-DC, 32 MHz crystal), PE4259 switch, U.FL

Topology per Semtech AN1200.40 Rev 1.1 Fig.5 (E428) / E449: VDD_IN+VBAT+VBAT_IO on 3V3_MAIN, 15uH DCC_SW-VREG (DC-DC), 470nF/100nF/10uF supply, DIO2 -> 100R/1nF -> PE4259 CTRL.
*** ALL TX match/harmonic filter and RX balun VALUES ARE TBD: AN1200.40 prints designators only. Take exact values from the native E449 regional BOM and re-tune on this stackup (AGENTS rule 5). ***
32 MHz crystal MPN TBD (CL 10 pF, ESR <=60R per SX1261/2 D5 Rev 1.2). DIO3 left open (TCXO fallback). No TX before a regional configuration (AGENTS rule 14).



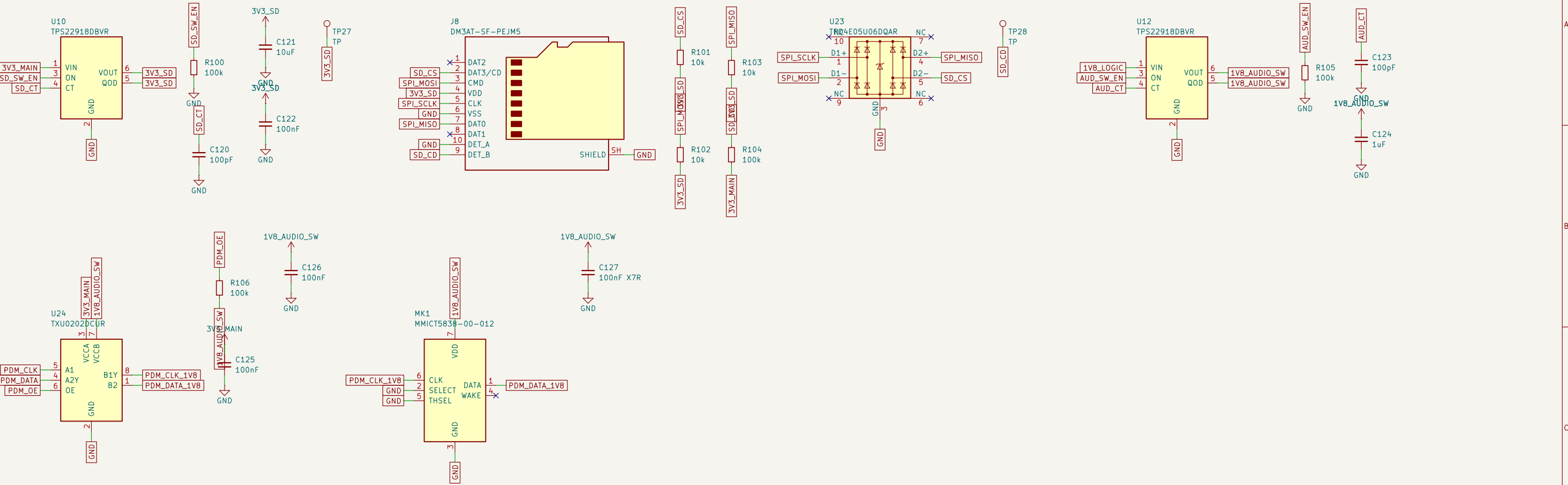
09 1090 MHz chain (BLB01 x2, TA2003A x2, ADL5513, MCP6566) and RP2040 capture

Chain per docs/ADSB_VALIDATION.md. BLB01 bias per BeRex rev 6.6 evaluation BOM (C1 100pF in, C2 12pF out, R1 6.8k/C3 12pF/L1 27nH Ven, R2 0R/C4 100pF/L2 82nH Vd). ADL5513 measurement mode (VSET=VOUT), 52.3R input termination, 47nF coupling, TADJ resistor and CLPF (1 nF basic value is too slow for 0.5 us pulses) are TBD/bench items. MCP6566T-E/OT base pin map, 2.2k pull-up to the RP2040 bank: threshold divider R86/R87 BLOCKED by OWN-09 (value TBD), R88/C88 DNP option: RP2040 PWM threshold, RP2040 per DIG-01 (Raspberry Pi minimal design): 12 MHz ABM8-272-T3 + 1k XOUT, 15pF loads; QSPL_S5 1k to USB_BOOT pad; RUN 10k/100nF + TC49535 P14, Domain 3V3_ADSB switched (P13). UART0: GP100 TX -> RP_UART_RX, GPIO1 RX <- RP_UART_TX, GPIO2 CTS <- RP_UART_RTS, GPIO3 RTS -> RP_UART_CTS (one-shot pin choice). Back-power of ESP32-driven lines into an OFF RP2040 is an open SYS-02 item.



10 microSD (DM3AT) on switched 3V3_SD; T5838 PDM mic on 1V8_AUDIO_SW

DIG-02: SPI mode (DAT3 = SD_CS), 10k pull-ups on CMD/DAT0/DAT3 to 3V3_SD (never 3V3_MAIN), no CLK pull-up, card detect 100k to 3V3_MAIN on TCA P00, TPD4E05U06 at the socket.
AUD-01/AUD-02: 1V8_AUDIO_SW = TPS22918 branch from 1V8_LOGIC (AUD_SW_EN, pull-down). TXU0202 A1->B1Y clock, B2->A2Y data, OE 100k to 1V8_AUDIO_SW.
T5838 SELECT=GND (DATA1/right), THSEL=GND and WAKE open (no AAD). 0.8 mm sound hole in the footprint, no paste on the port.



11 Mounting holes, fiducials

Three M2.5 mounting holes in the left strip (the 2S holder covers the back for x > 14.7 mm; holder is fixed to the rear shell); three fiducials.

- H1

M2.5
- H2

M2.5
- H3

M2.5
- FID1

Fiducial
- FID2

Fiducial
- FID3

Fiducial

CANDIDATE – NOT FOR MANUFACTURE – see README gates		
StratosCore (CERN–OHL–P–2.0) – Claude one-shot candidate		
Sheet: /11 Mechanical/ File: 11_mechanical.kicad_sch		
Title: 11 Mounting holes, fiducials		
Size: A3	Date:	Rev: A-claude-oneshot-1
KiCad E.D.A. 10.0.6		Id: 12/12